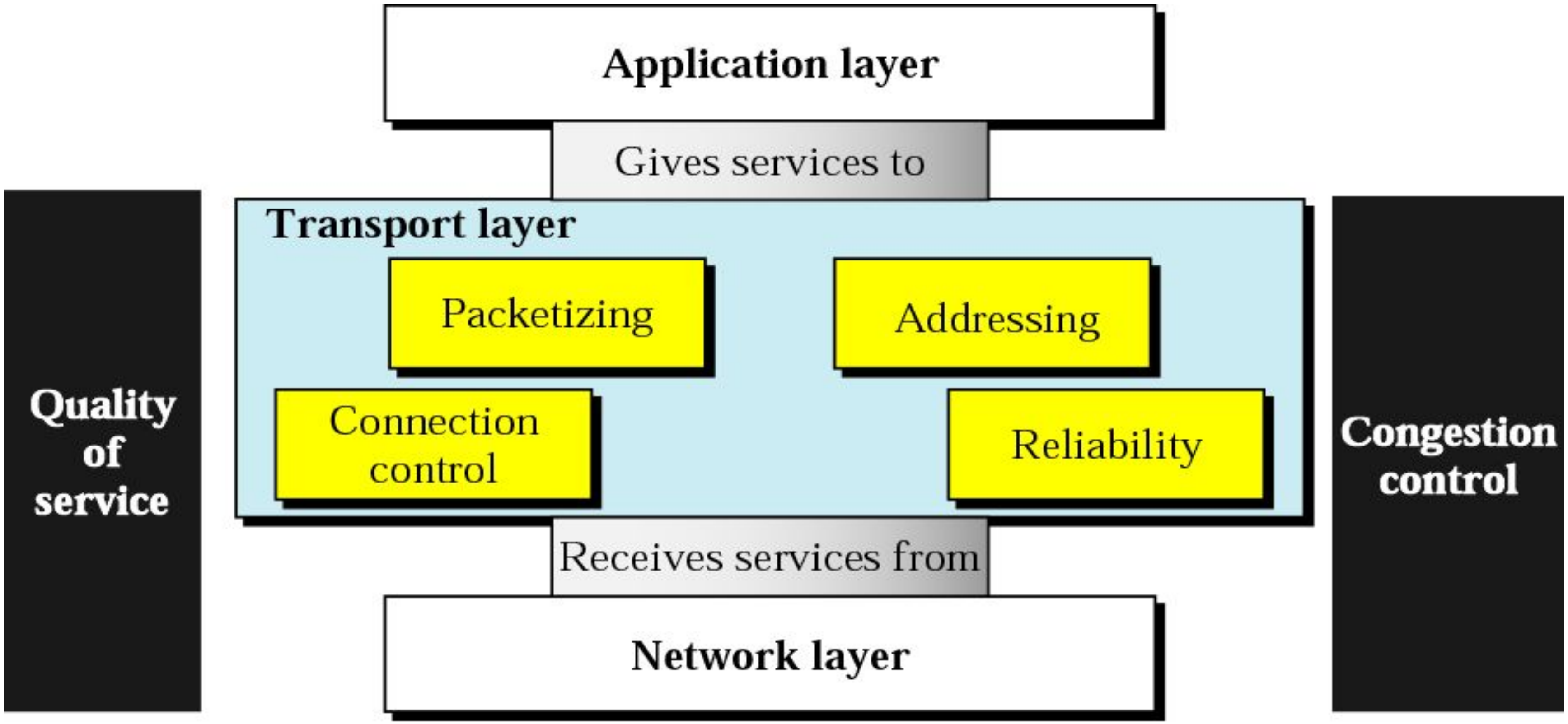
Computer Communication & Networks

Lecture 25

Transport Layer: Congestion Control

Transport Layer



Transport Layer Topics to Cover

- *UDP, TCP, STCP*
- ***Congestion Control***

Congestion

- Congestion in a network may occur if the load on the network—the number of packets sent to the network—is greater than the capacity of the network—the number of packets a network can handle. Congestion control refers to the mechanisms and techniques to control the congestion and keep the load below the capacity.

Congestion in packet switching networks

Question:

Is it possible to avoid/reduce congestion by increasing buffer size?

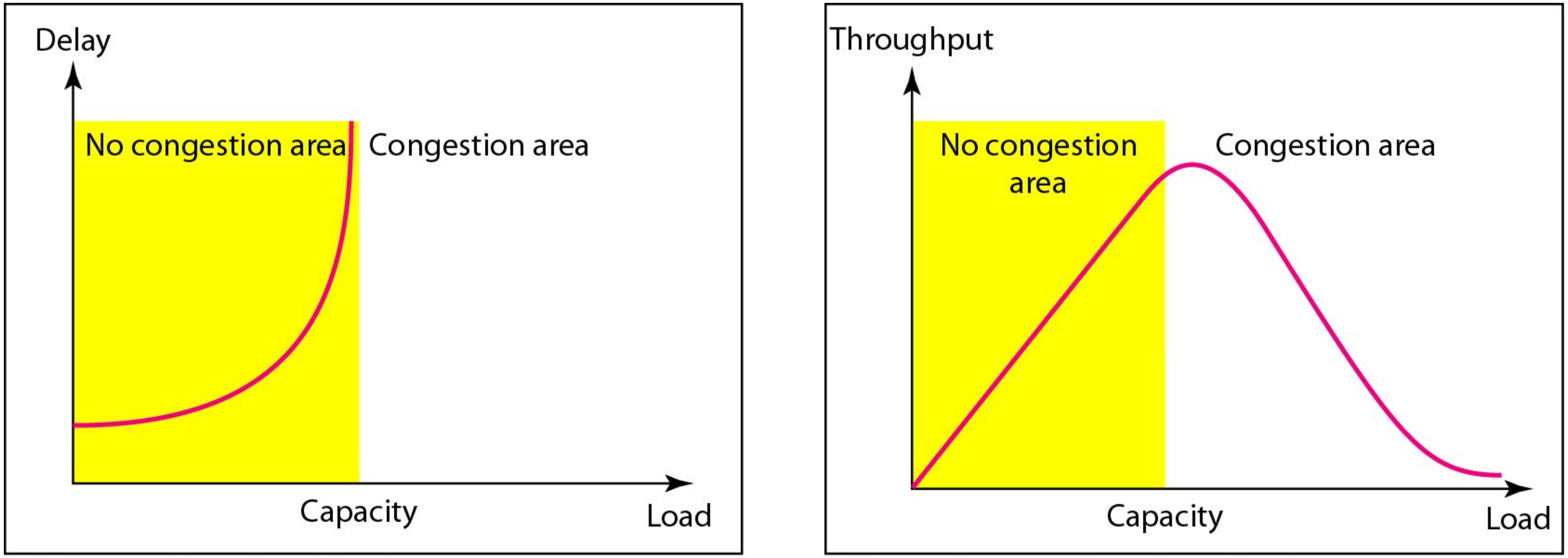
Congestion in packet switching networks

NO

Buffer is larger, then waiting time is longer. It will cause time out and retransmit packets, thus more congestion. If buffer size is infinite, then packet can delay forever.

Congestion control is a very hard problem

Packet delay and throughput as functions of load



a. Delay as a function of load

b. Throughput as a function of load

Congestion Control

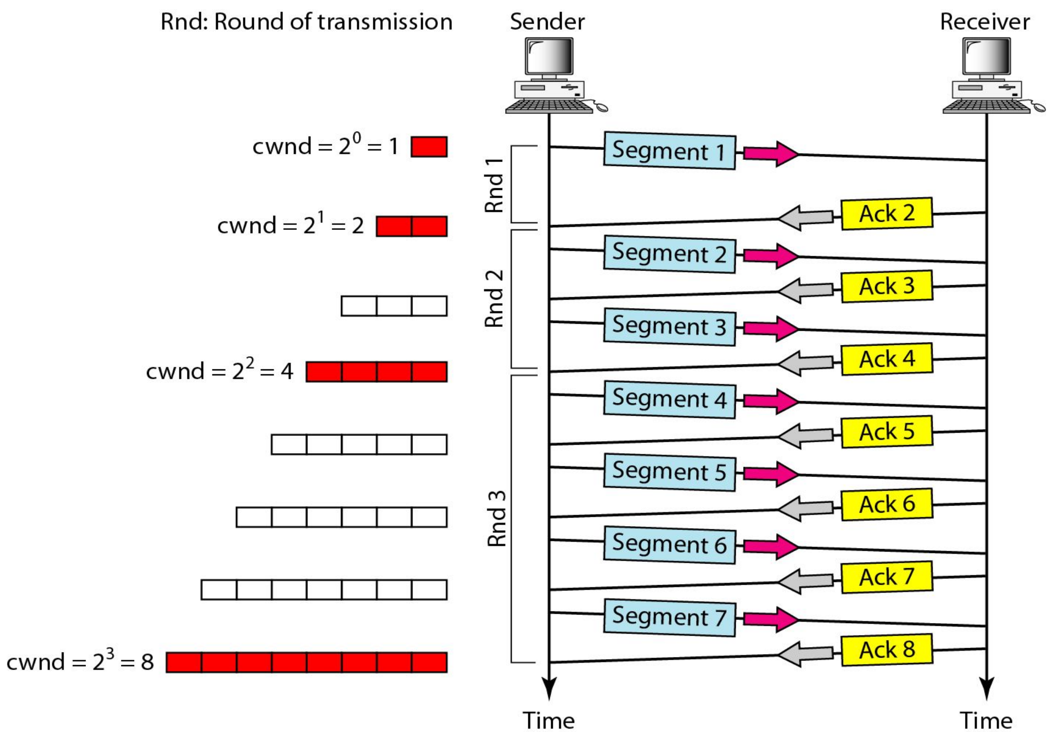
- Congestion control refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened. In general, we can divide congestion control mechanisms into two broad categories:
  - open-loop congestion control (prevention)
  - closed-loop congestion control (removal)

Example

- To better understand the concept of congestion control, let us give an example: Congestion Control in TCP.

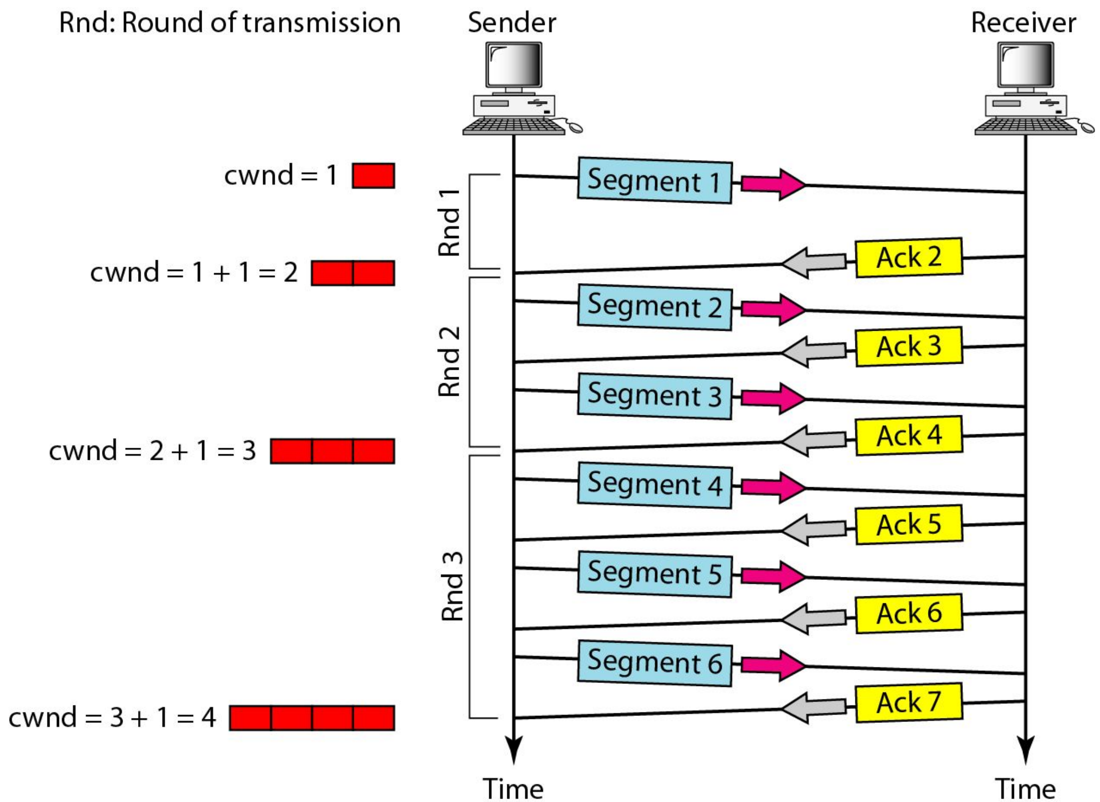
Slow Start, Exponential Increase

- In the slow-start algorithm, the size of the congestion window increases exponentially until it reaches a threshold.



Congestion Avoidance: Additive Increase

- In the congestion avoidance algorithm, the size of the congestion window increases additively until congestion is detected.



Congestion Example